

# Electronics | Lecture # 01 | ECAT | MDCAT | Bilal Zia

# REC → CIR → CHAP-WISE → PDF ↗ Lec #01 → Play It

Metalloids  
CARBON family (IVA)

# CONDUCTOR

↘  
Metal

# INSULATOR

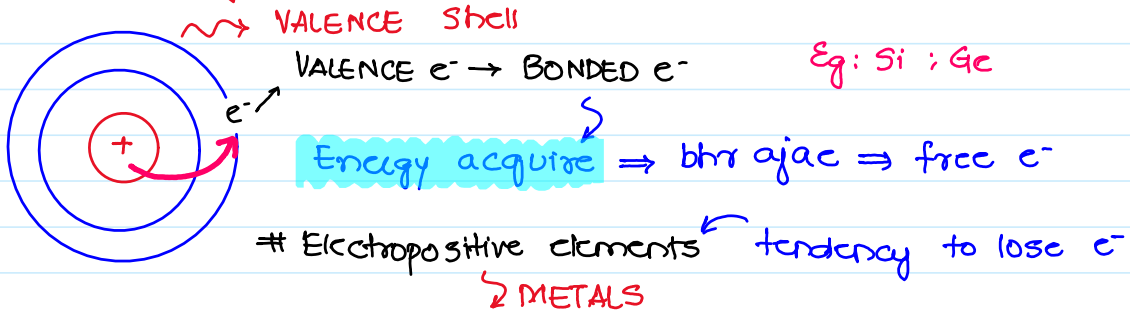
↘  
Non-Metal

# SEMI-CONDUCTOR

PARTIAL CONDUCTORS

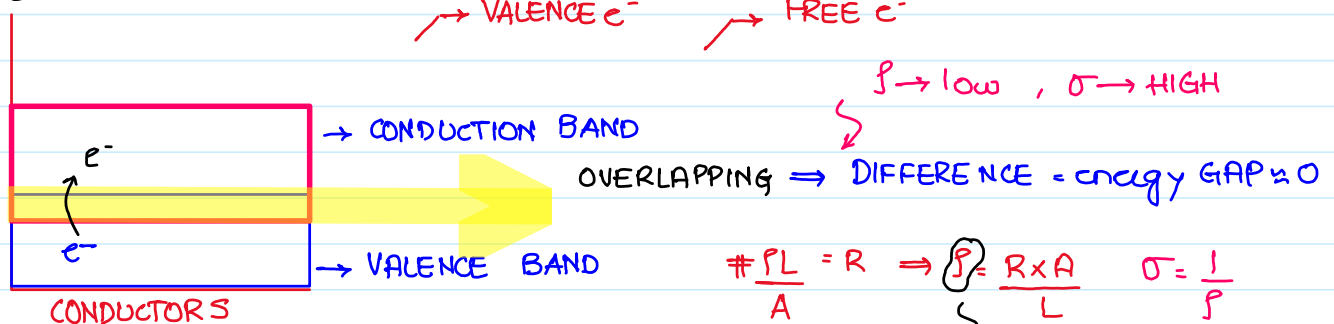
CONDUCTIVITY: CONTROL

# Band Theory of Solids



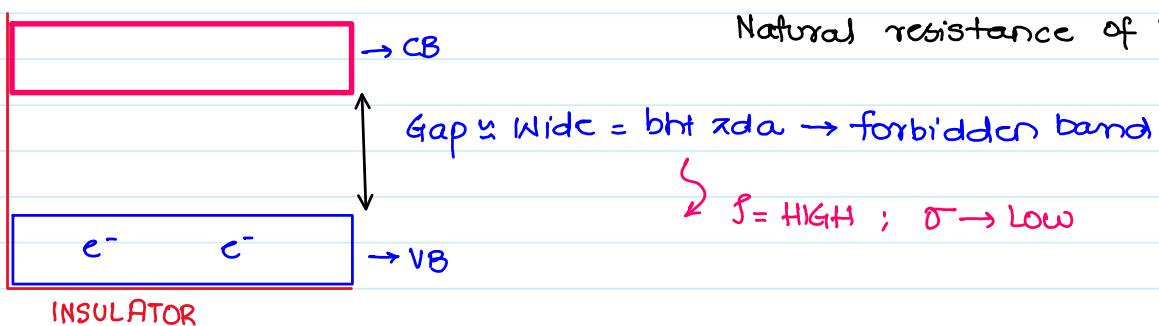
Eg: Si ; Ge

ENERGY

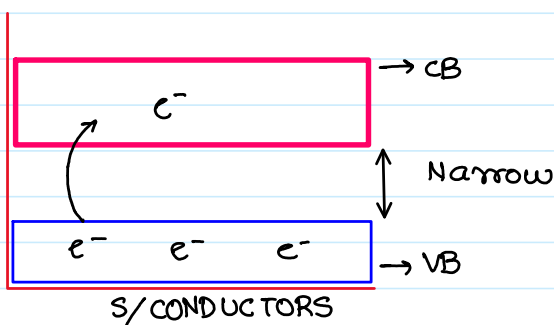


Natural resistance of Material

E



E



CONDUCTIVITY:

$$\sigma_I < \sigma_S < \sigma_C \quad \checkmark$$

RESISTIVITY:

$$\rho_C < \rho_S < \rho_I$$

# SC  $\rightarrow$  Group IVA  $\left\{ \begin{array}{l} \text{Si} \Rightarrow 4e^- \\ \text{Ge} \Rightarrow 4e^- \end{array} \right.$

# Pure SC crystals  $\rightarrow$  INTRINSIC S/C

# SC  $\rightarrow$  Group IVA  $\begin{cases} 31 \Rightarrow 7e^- \\ 4e^- \Rightarrow 4e^- \end{cases}$

# Pure SC crystals  $\rightarrow$  INTRINSIC S/C

$4e^-$  in Valence shell

Impurity ko add krte hn. **DOPING**

\* Si + IMPURITY  $\rightarrow$  doping

# doping krne k bd  $\rightarrow$  EXTRINSIC S/C  
 $\rightarrow$  DOPED S/C

to inc. the conductivity of S/C

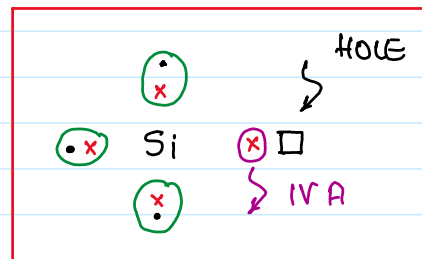
## ① TRIVALENT DOPING

PURE S/C + IMPURITY  
 $\rightarrow$  S/C

Group IIIA (B, Al, Indium)

IIIA Imp  
 $\rightarrow$  Valence ( $3e^-$ )

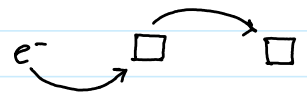
- PTYPE  
- ACCEPTOR



$\rightarrow$  VACANCY

$\rightarrow$  jha pr  $e^-$  askta

PTYPE S/C



#  $L = mvr = n \times \frac{h}{2\pi}$

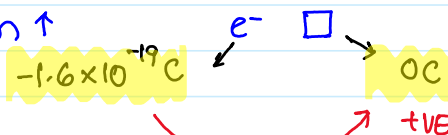
# Mobility of free  $e^-$

# Pure S/C

# additional HOLE agya per 'Si' atom

free  $e^-$  Holes

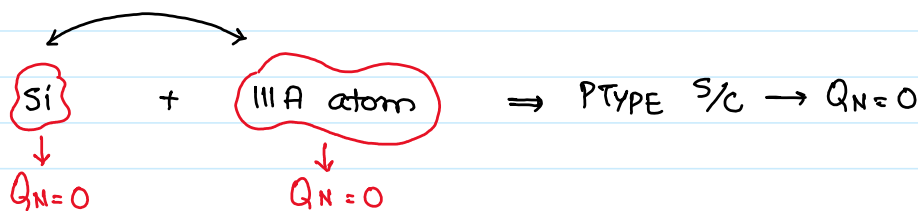
# Hole mobilization  $\uparrow$



# MAJORITY : HOLES

# MINORITY :  $e^-$

# P Type S/C  $\rightarrow$  overall electrically  $\rightarrow$  **Neutral** Ya +ve charge



## ② PENTAVALENT DOPING

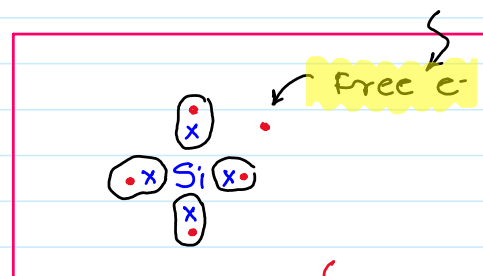
Si/Ge + Impurity

Group VA

$5e^-$  in VS

- N Type doping

DONOR doping



S/C  $\begin{cases} \text{free } e^- + \text{CONT.} \\ \text{Holes} \end{cases}$

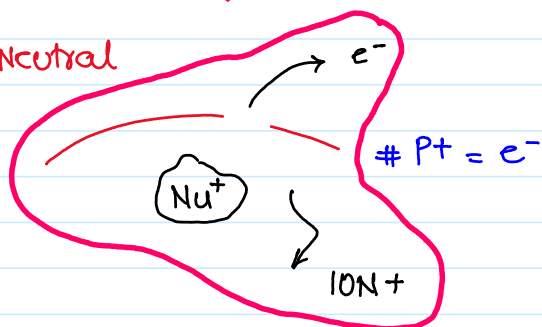
$P^+ = e^-$

Majority:  $e^-$   
Minority: Holes

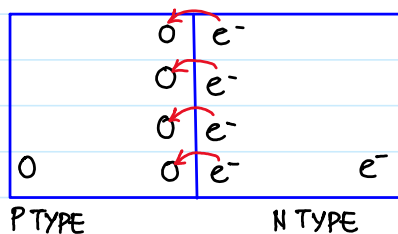
Electrically Neutral

Majority:  $e^-$   
 Minority: Holes

Electrically Neutral



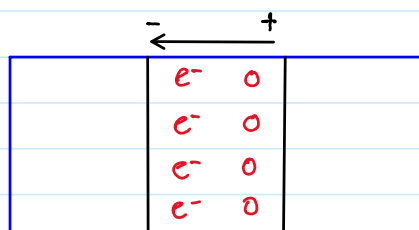
# PN JUNCTION DIODE formation:



# external battery = NIL  
 # ext. PD = Not applied



#  $E = \frac{\Delta V}{L}$



DIODE

Si = 0.7V

Ge = 0.3 ~ 0.4V

$\Delta V$  DEPLETION region : Potential barrier